Track: DEPI Round 5 (Machine Learning)

# Kemet Hieroglyphic Studio

## PROJECT PLANNING REPORT

Kemet Hieroglyphic Studio: Executive Project Plan & Milestone Roadmap

## STAKEHOLDER ANALYSIS REPORT

Kemet Hieroglyphic Studio: Stakeholder Mapping, Engagement & Value Delivery



Machine Learning Track

# 1. Project Overview & Strategic Objectives

Project Goal: Create an easy-to-use web platform where users can identify ancient Egyptian hieroglyphs from images, learn their historical meanings, and listen to clear spoken pronunciations.

**Core Feature**

## Visual Detection

**What It Does for the User**

Users upload photos and use an interactive cropping tool to identify hieroglyphs.

**Key Deliverable**

Interactive image cropper and detection model.

## Symbol Catalog

**What It Does for the User**

Provides clear descriptions, Gardiner codes, and cultural background for signs.

**Key Deliverable**

Searchable catalog of 179 curated hieroglyphs.

## AI Scribe Chat

**What It Does for the User**

Answers user questions about hieroglyphic symbols and history in English and Arabic.

**Key Deliverable**

Conversational assistant using authentic hieroglyphs.

## Voice Studio

**What It Does for the User**

Plays natural spoken pronunciations and audio stories for catalog symbols.

**Key Deliverable**

Fast text-to-speech audio player.

# 2. Scope Definition: In-Scope vs. Out-of-Scope

**Project Area**

## Image Tools

**In-Scope Deliverables (Current Release)**

- Photo upload (JPG/PNG/WebP) and clipboard paste.
- Interactive canvas cropping tool to isolate single signs.
- Automatic screening check for hieroglyphic content.
- Bounding box overlay showing detection confidence scores.

**Out-of-Scope / Future Enhancements**

- Translating full sentences of continuous text.
- 3D scanning or photogrammetry of temple walls.
- Automatic reconstruction of broken or eroded stones.

## Chat & Text

**In-Scope Deliverables (Current Release)**

- Conversational questions and answers in English and Arabic.
- Real Unicode hieroglyphic character display.
- Basic symbol types (phonograms, ideograms, determinatives).

**Out-of-Scope / Future Enhancements**

- Complex grammar analysis of long ancient papyri.
- Storing private user chat logs in a database.
- User registration, passwords, and personal accounts.

## Catalog & Audio

**In-Scope Deliverables (Current Release)**

- Browsing symbols by standard Gardiner categories.
- Symbol descriptions, phonetic values, and transliteration.
- On-demand audio playback for catalog items.

**Out-of-Scope / Future Enhancements**

- Microphone voice-to-text input.
- Selecting different character voice actors.
- Audio file downloads and podcast feeds.

# 3. Project Schedule & Milestone Roadmap

| Sprint / Phase | Main Objective & Work Completed | Delivered Milestone | Status |
|---|---|---|---|
| **Sprint 1 (Weeks 1–2)** | Collected project requirements and cleaned the 179-symbol hieroglyphic catalog. | Verified sign dictionary and baseline metadata. | Completed |
| **Sprint 2 (Weeks 3–4)** | Trained the symbol detection model on dataset and evaluated accuracy. | Trained model weights (best.pt) ready for inference. | Completed |
| **Sprint 3 (Weeks 5–6)** | Built FastAPI endpoints and added initial image check to filter invalid photos. | Fast backend detection API running locally. | Completed |
| **Sprint 4 (Weeks 7–8)** | Connected conversational AI for sign questions and set up text-to-speech audio. | AI Scribe Chat and Voice Studio functional. | Completed |
| **Sprint 5 (Weeks 9–10)** | Created modern single-page interface and added interactive symbol cropper. | Responsive web interface connected to backend. | Completed |
| **Sprint 6 (Weeks 11–12)** | Ran 11 automated pytest test cases and fixed responsive layout on laptop screens. | Final platform tested, polished, and documented. | Completed |

Project Sign-Off: Kemet Hieroglyphic Studio Team | Digital Egypt Pioneers Initiative (DEPI Round 5) | Machine Learning Track

# 1. Stakeholder Identification & Classification

Overview: Kemet Studio connects learners, academic mentors, domain experts, and engineers to create an accessible tool for exploring ancient Egyptian hieroglyphs.

| Stakeholder Group | Key Personas & Entities | Primary Motivation & Expectations | Project Role |
|---|---|---|---|
| **Primary Learners** | Students, museum visitors, history enthusiasts, and amateur learners. | Quickly crop and identify symbols from photos without requiring prior language knowledge. | **End Users** |
| **Academic Mentors** | DEPI evaluation committee, project mentors, and academic supervisors. | Verify model accuracy, clear software architecture, and complete project documentation. | **Evaluators** |
| **Domain Specialists** | Egyptologists, museum educators, and cultural heritage researchers. | Ensure historical accuracy in sign classifications, Gardiner codes, and Unicode glyph rendering. | **Domain Advisors** |
| **Engineering Team** | Machine learning engineer, full-stack web developer, and QA tester. | Build a stable, high-performance web platform meeting graduation project requirements. | **Developers** |

# 2. Power-Interest Matrix & Engagement Strategy

Power / Interest

Quadrant

## Keep Satisfied

**High / Low**

Stakeholder Group

**DEPI Leadership & Technical Coordinators**

Management & Engagement Strategy

Ensure on-time completion, stable local setup without complex steps, and clean code.

Quadrant

## Manage Closely

**High / High**

Stakeholder Group

**DEPI Evaluators & Academic Mentors**

Management & Engagement Strategy

Provide regular progress demos, pass automated test suites, and present clear methodology.

Quadrant

## Monitor

**Low / Low**

Stakeholder Group

**General Public & Casual Web Explorers**

Management & Engagement Strategy

Maintain high browser availability, responsive layouts, and transparent AI disclaimers.

Quadrant

## Keep Informed

**Low / High**

Stakeholder Group

**Students, Museum Visitors & History Fans**

Management & Engagement Strategy

Offer an easy web interface, visual guides, preset examples, and instant feedback.

# 3. Stakeholder Value Proposition & Needs Mapping

**Stakeholder Group**

## Beginners & Students

| Main User Need / Challenge | Solution Provided by Kemet Studio | Success Measure |
|---|---|---|
| Hieroglyphic signs look complex and hard to search. | Interactive image cropping tool with instant YOLOv11 symbol recognition. | Successful sign identification within 2 seconds. |

## History Enthusiasts

| Main User Need / Challenge | Solution Provided by Kemet Studio | Success Measure |
|---|---|---|
| Lack of accurate pronunciations and cultural context. | Voice Studio with clear spoken audio and symbol storytelling in English & Arabic. | Reliable audio playback across all catalog entries. |

## Academic Evaluators

| Main User Need / Challenge | Solution Provided by Kemet Studio | Success Measure |
|---|---|---|
| Require evidence of machine learning rigour and reliability. | YOLOv11 trained on 15,000 images, standard Gardiner labels, and 11 unit tests. | 100% automated test pass rate and high detection accuracy. |

## Domain Specialists

| Main User Need / Challenge | Solution Provided by Kemet Studio | Success Measure |
|---|---|---|
| Prevent inaccurate translations and fake glyph shapes. | Curated 179-sign dictionary mapped directly to authentic Unicode characters. | Zero fictional characters; valid Gardiner classifications. |

# 4. Communication & Governance Plan

| Cadence | Channel / Activity | Target Audience | Key Objective & Deliverable |
|---|---|---|---|
| Bi-weekly | **Sprint Reviews & Demos** | **Mentors & Evaluators** | Demonstrate working detection, chat, and voice features against milestone plan. |
| Milestone releases | **User Testing Sessions** | **Students & Learners** | Test image cropping accuracy, layout readability, and chat responses. |
| Continuous / Final | **Codebase & Documentation** | **Technical Evaluators** | Share clean GitHub repository, setup guide, dependency list, and test scripts. |

Project Sign-Off: Kemet Hieroglyphic Studio Team | Digital Egypt Pioneers Initiative (DEPI Round 5) | Machine Learning Track